

LifeLineX Project Documentation

Phase 3: Data Modeling & Relationships

1. Objective

The objective of Phase 3 is to design and implement the **data model** of the LifeLineX system. This includes:

- Creating **custom objects** for all key entities.
- Adding **fields** to capture detailed information.
- Establishing **relationships** between objects.
- Designing **page layouts** and **compact layouts** for user experience.
- Validating the model in **Schema Builder**.

This phase ensures the system is capable of representing real-world organ donation and transportation workflows.

2. Steps Implemented

Step 3.1: Planning the Data Model

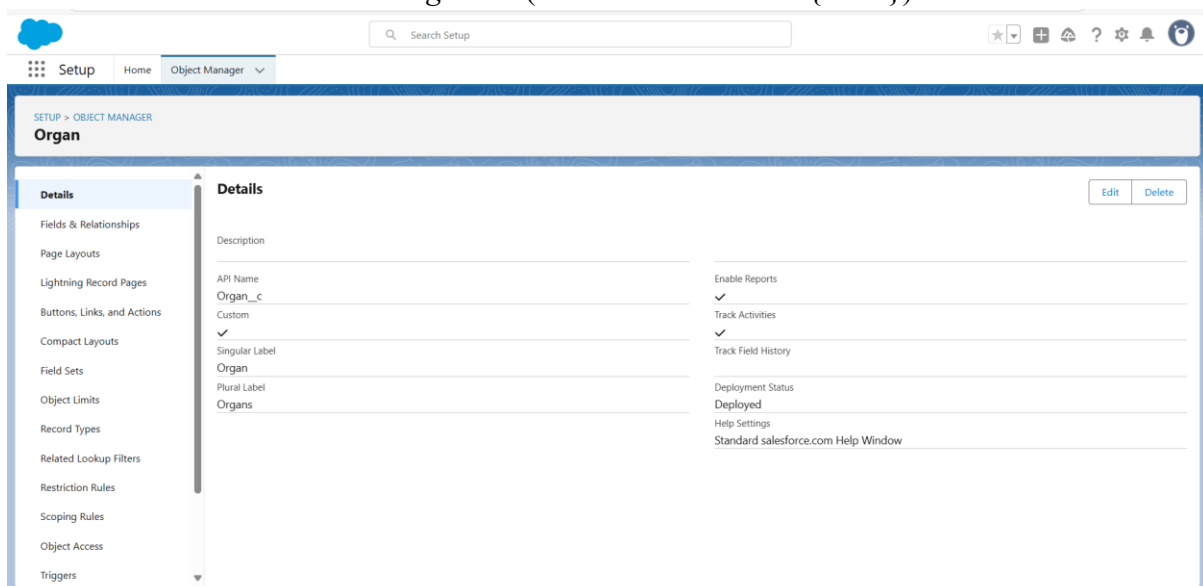
- Identified entities required in LifeLineX:
 - **Organ**
 - **Transport Case**
 - **Doctor**
 - **Courier**
 - **Alerts**
- Defined fields and relationships for each entity.

Step 3.2: Creating Custom Objects

Created the following **custom objects** in *Setup* → *Object Manager* → *New Object*:

1. Organ

- Record Name: *Organ ID* (Auto Number: ORG-{0000})



2. Transport Case

- Record Name: *Case Number* (Auto Number: CASE-{0000})

The screenshot shows the Salesforce Setup interface for the 'Transport Case' object. The left sidebar contains a navigation menu with options like 'Details', 'Fields & Relationships', 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Compact Layouts', 'Field Sets', 'Object Limits', 'Record Types', 'Related Lookup Filters', 'Restriction Rules', 'Scoping Rules', 'Object Access', and 'Triggers'. The 'Details' section is active, showing fields for 'Description', 'API Name' (Transport_Case_c), 'Custom' (checked), 'Singular Label' (Transport Case), and 'Plural Label' (Transport Cases). On the right, there are checkboxes for 'Enable Reports' (checked), 'Track Activities' (checked), and 'Track Field History'. Below these are 'Deployment Status' (Deployed), 'Help Settings', and a link to 'Standard salesforce.com Help Window'. 'Edit' and 'Delete' buttons are in the top right corner.

https://orgfarm-28f61742ad-dev-ed.develop.lightning.force.com/one/one.app#/setup/ObjectManager/01gK000002Pymw/PageLayouts/View

3. Doctor

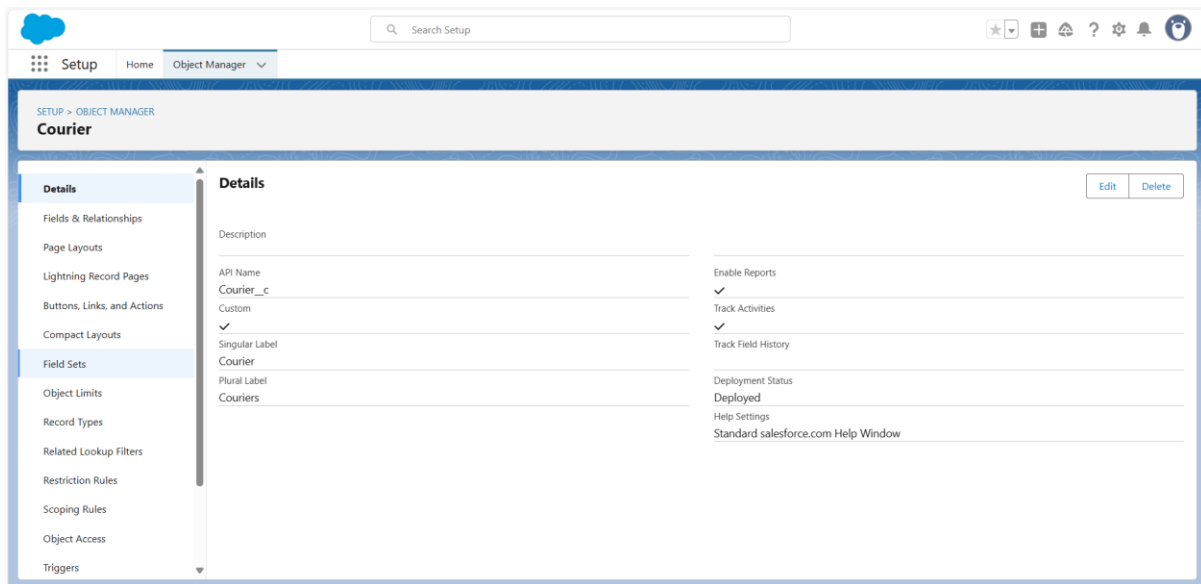
- Record Name: *Doctor ID* (Auto Number: DOC-{0000})

The screenshot shows the Salesforce Setup interface for the 'Doctor' object. The left sidebar is identical to the previous screenshot. The 'Details' section is active, showing fields for 'Description', 'API Name' (Doctor_c), 'Custom' (checked), 'Singular Label' (Doctor), and 'Plural Label' (Doctors). On the right, there are checkboxes for 'Enable Reports' (checked), 'Track Activities' (checked), and 'Track Field History'. Below these are 'Deployment Status' (Deployed), 'Help Settings', and a link to 'Standard salesforce.com Help Window'. 'Edit' and 'Delete' buttons are in the top right corner.

https://orgfarm-28f61742ad-dev-ed.develop.lightning.force.com/one/one.app#/setup/ObjectManager/01gK000002Pymw/PageLayouts/View

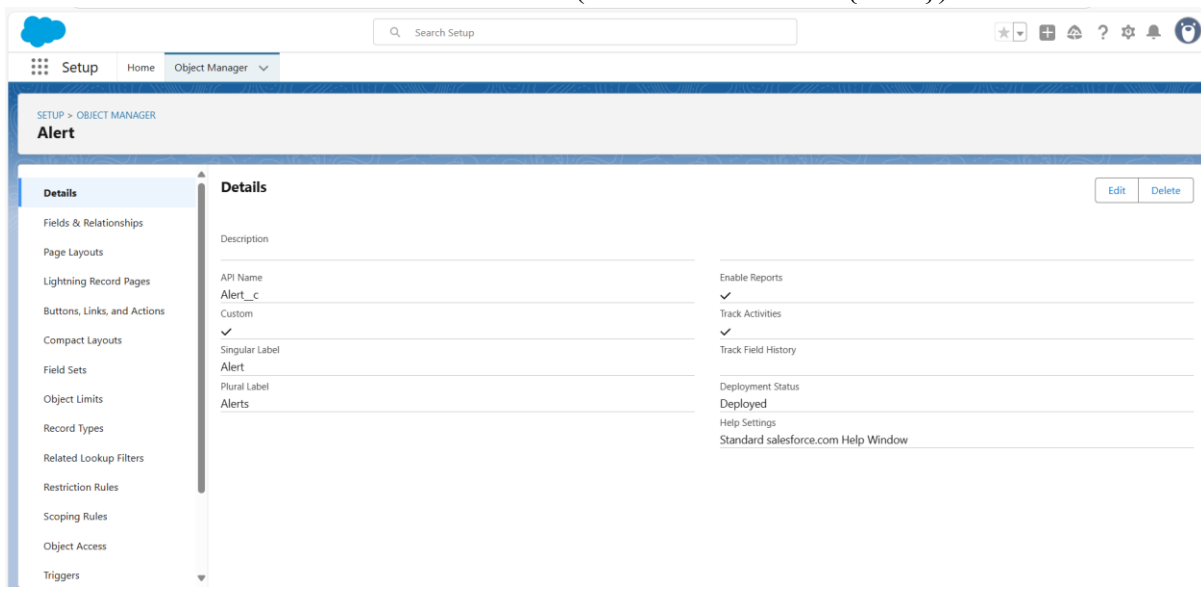
4. Courier

- Record Name: *Courier ID* (Auto Number: CR-{0000})



5. Alerts

- Record Name: *Alert Number* (Auto Number: ALT- $\{0000\}$)



Step 3.3: Adding Fields

Organ Object

- Organ Type → Picklist (Heart, Kidney, Liver, Lung, Pancreas)
- Donor Hospital → Text
- Condition → Picklist (Good, Fair, Critical)
- Availability Status → Picklist (Available, Assigned, Transported, Used)

SETUP > OBJECT MANAGER

Organ

Details

Fields & Relationships
6 Items, Sorted by Field Label

Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Availability Status	Availability_Status__c	Picklist		
Condition	Condition__c	Picklist		
Created By	CreatedById	Lookup(User)		
Donor Hospital	Donor_Hospital__c	Text(40)		
Last Modified By	LastModifiedById	Lookup(User)		
Organ ID	Name	Auto Number		✓
Organ Type	Organ_Type__c	Picklist		
Owner	OwnerId	Lookup(User,Group)		✓

Transport Case Object

- Source Hospital → Text
- Destination Hospital → Text
- Status → Picklist (Pending, In Transit, Delivered, Cancelled)
- Pickup Time → Date/Time
- Delivery Time → Date/Time
- Courier → Lookup (Courier)
- Organ → Lookup (Organ)

SETUP > OBJECT MANAGER

Transport Case

Details

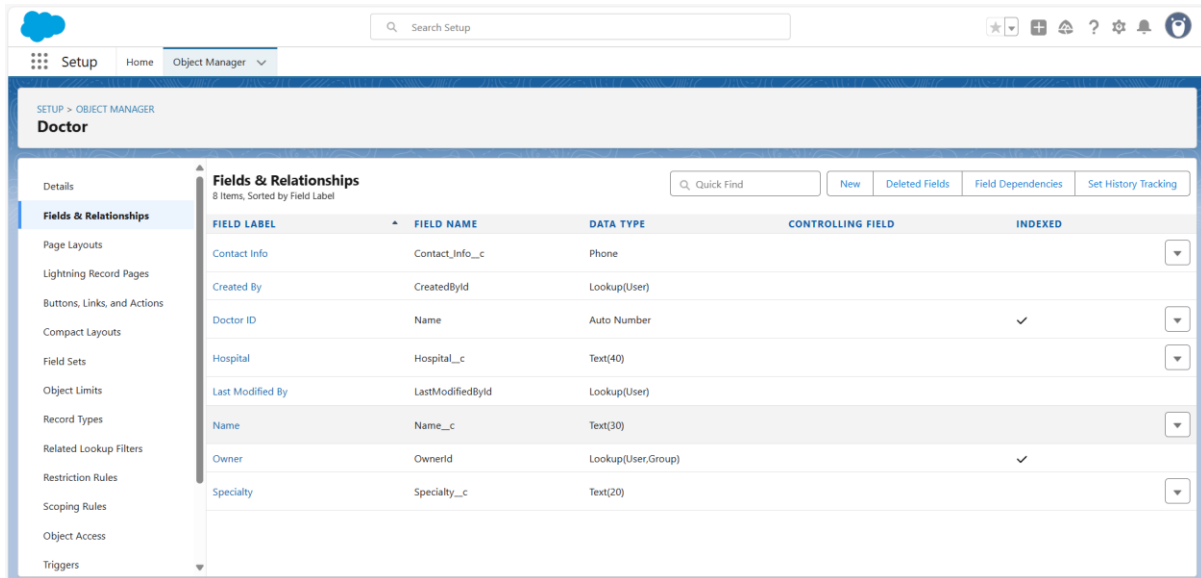
Fields & Relationships
11 Items, Sorted by Field Label

Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Courier	Courier__c	Lookup(Courier)		✓
Created By	CreatedById	Lookup(User)		
Delivery Time	Delivery_Time__c	Date/Time		
Destination Hospital	Destination_Hospital__c	Text(40)		
Last Modified By	LastModifiedById	Lookup(User)		
Organ	Organ__c	Lookup(Organ)		✓
Owner	OwnerId	Lookup(User,Group)		✓
Pickup Time	Pickup_Time__c	Date/Time		
Source Hospital	Source_Hospital__c	Text(40)		
Status	Status__c	Picklist		

Doctor Object

- Name → Text
- Specialty → Text
- Hospital → Text
- Contact Info → Phone / Email



SETUP > OBJECT MANAGER

Doctor

Details

Fields & Relationships
8 Items, Sorted by Field Label

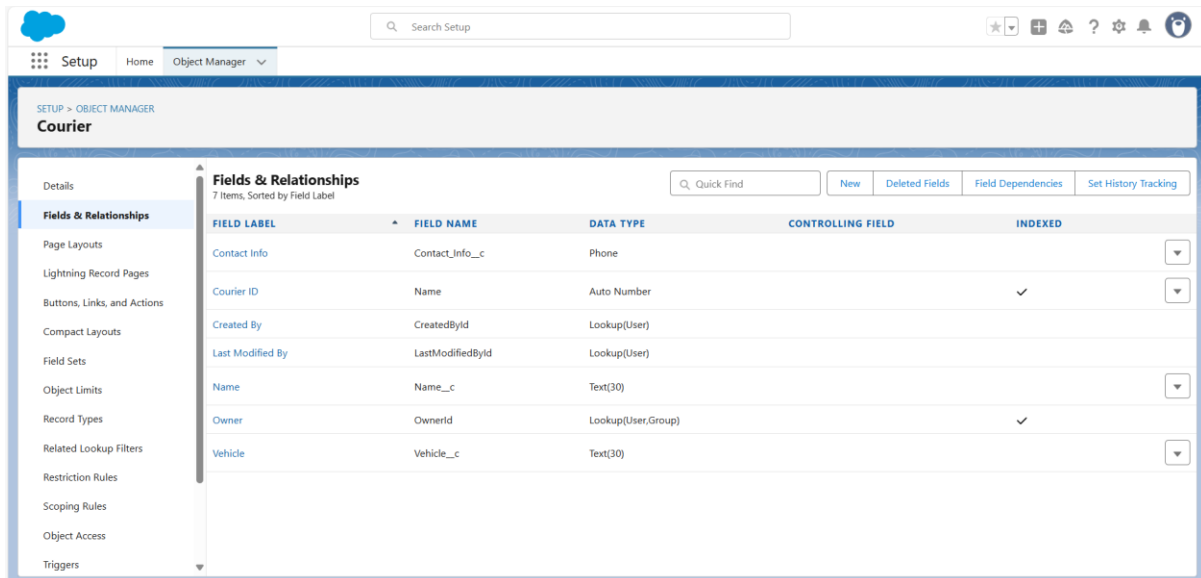
Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Contact Info	Contact_Info__c	Phone		
Created By	CreatedById	Lookup(User)		
Doctor ID	Name	Auto Number		✓
Hospital	Hospital__c	Text(40)		
Last Modified By	LastModifiedById	Lookup(User)		
Name	Name__c	Text(30)		
Owner	OwnerId	Lookup(User,Group)		✓
Specialty	Specialty__c	Text(20)		

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers

Courier Object

- Name → Text
- Vehicle → Text
- Contact Info → Phone / Email



SETUP > OBJECT MANAGER

Courier

Details

Fields & Relationships
7 Items, Sorted by Field Label

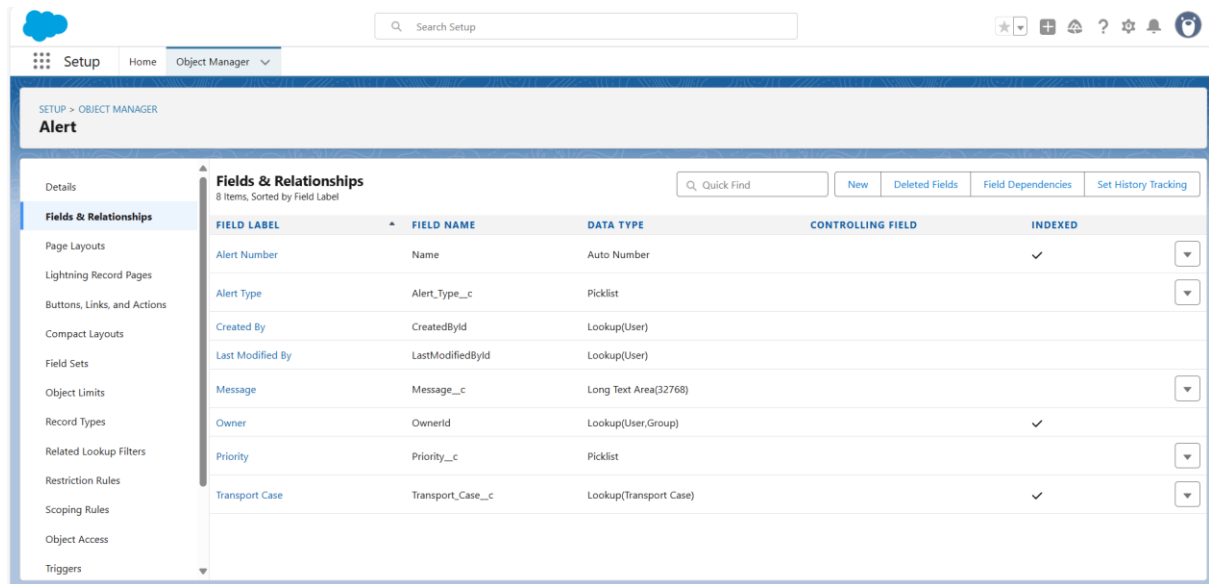
Q Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Contact Info	Contact_Info__c	Phone		
Courier ID	Name	Auto Number		✓
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Name	Name__c	Text(30)		
Owner	OwnerId	Lookup(User,Group)		✓
Vehicle	Vehicle__c	Text(30)		

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers

Alerts Object

- Alert Type → Picklist (Delay, Emergency, Delivered, General)
- Message → Long Text Area
- Priority → Picklist (High, Medium, Low)
- Related Case → Lookup (Transport Case)



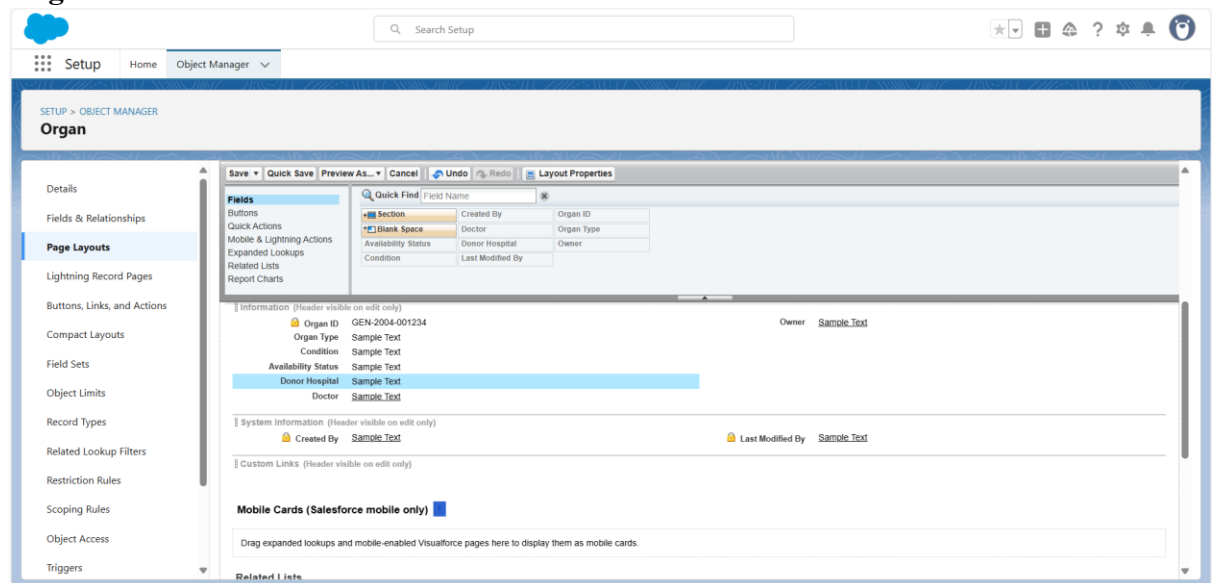
Step 3.4: Defining Relationships

- **Organ → Transport Case** : Lookup (one organ may belong to multiple cases over time)
- **Courier → Transport Case** : Lookup
- **Doctor → Organ** : Lookup (a doctor is responsible for an organ)
- **Alerts → Transport Case** : Master-Detail (alerts are dependent on the transport case)

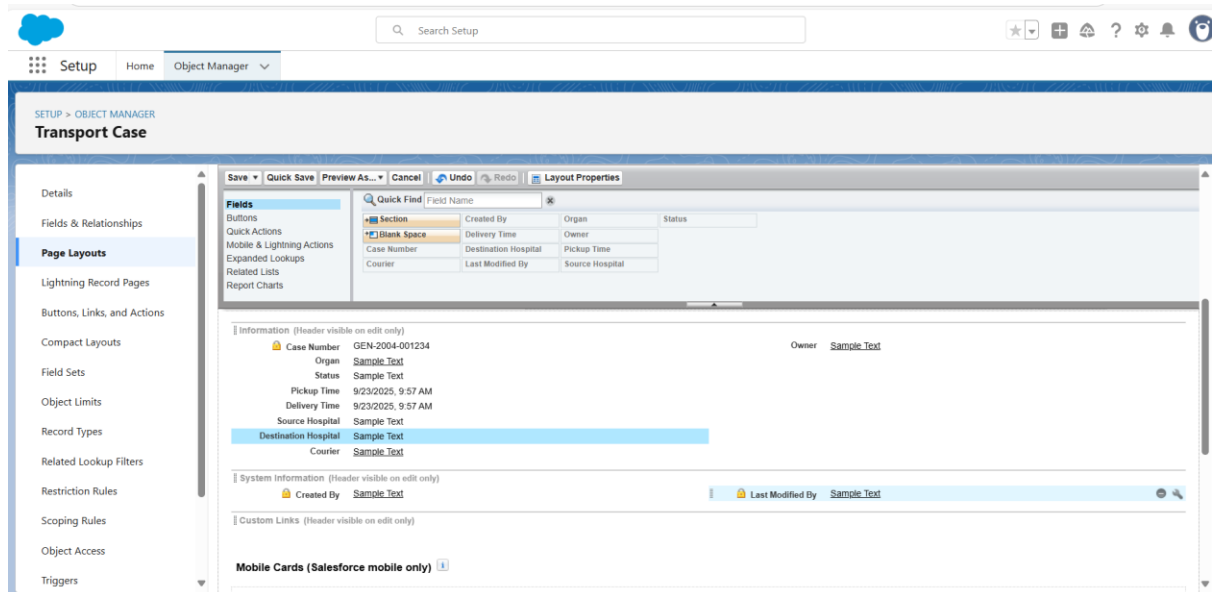
Step 3.5: Configuring Layouts

- Designed **Page Layouts** for each object to group fields logically.

Organ

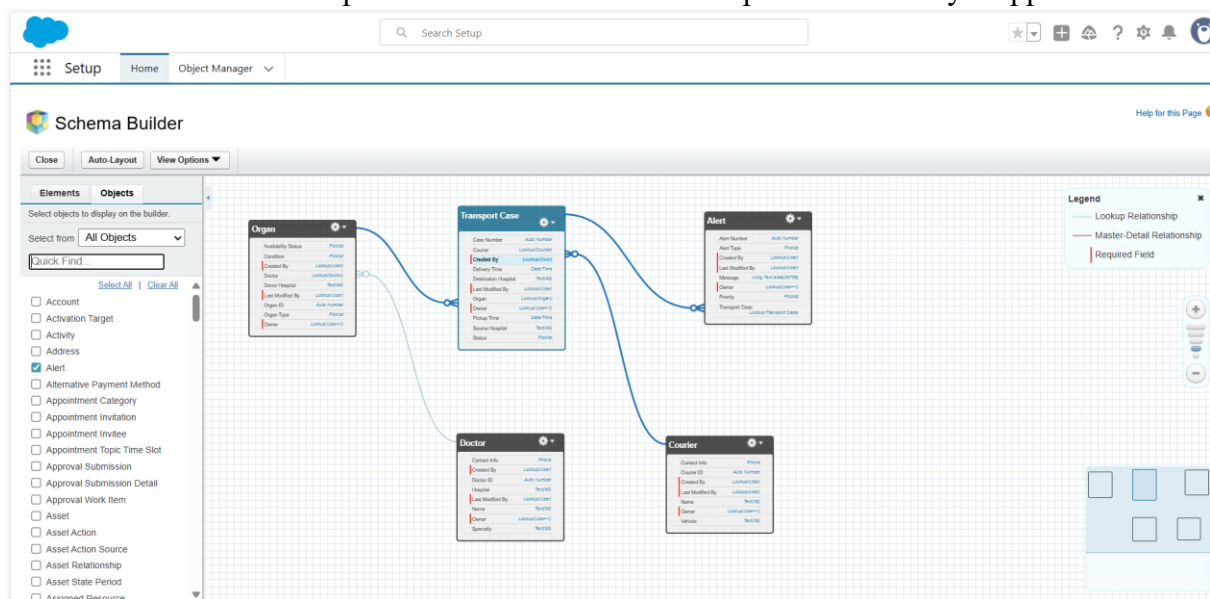


Transport Case



Step 3.6: Schema Builder Verification

- Opened Schema Builder from Setup.
- Added all custom objects.
- Verified that lookups and master-detail relationships were correctly mapped.



4. Dependencies

- **Object-level permissions (Profiles & Permission Sets)** → To be completed in Phase 4.
- **OWD (Organization-Wide Defaults)** → To be implemented in Phase 4.
- **Sharing Rules** → To be configured in Phase 4.